

✓ Homework 5: Model Evaluation

After going through the lecture notebooks, work on the homework below.

Question

Using the Breast Cancer Wisconsin dataset [1] prepared in Section 2, select a classification model based on `model_type`, build a pipeline with `StandardScaler()` and the classifier, evaluate it with `k`-fold cross-validation on the training data, and return the mean accuracy as a `float`.

Supported `model_type` values are:

- `'logistic_regression'`
- `'decision_tree'`
- `'svm'`

Inputs / Outputs:

```
homework(X_train: pd.DataFrame, y_train: pd.Series, model_type: str, k: int) -> float
```

Argument	Type	Description
<code>X_train</code>	<code>pd.DataFrame</code>	Training features from Section 2
<code>y_train</code>	<code>pd.Series</code>	Training labels from Section 2
<code>model_type</code>	<code>str</code>	Model name: <code>'logistic_regression'</code> , <code>'decision_tree'</code> , or <code>'svm'</code>
<code>k</code>	<code>int</code>	Number of folds for cross-validation
Return	Type	Description
<code>mean_score</code>	<code>float</code>	Mean cross-validation accuracy

Example:

```
result = homework(X_train, y_train, 'logistic_regression', 5)
print(result)
```

```
0.9802XXXXXXXXXXXX
```

Note:

- Submit **only** the `homework()` function to Omnicampus.
- Do **NOT** include `import` statements or data loading code in your submission.
- Return the mean accuracy as a `float`.

Deadline

Wed, May 27 (11:00 UTC)

✓ 1. Importing Libraries

```
# Please do not modify this cell
import numpy as np
import pandas as pd

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.svm import SVC
```

✓ 2. Loading Data

We use the **Breast Cancer Wisconsin** dataset [1] from scikit-learn. It contains 569 samples with 30 features for binary classification: benign (1) vs. malignant (0).

Note: Do NOT modify the following cell!

```
# Please do not modify this cell
data = load_breast_cancer(as_frame=True)
X = data.frame.drop(columns=['target'])
y = data.frame['target']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print(f"X_train shape: {X_train.shape}")
print(f"y_train value counts:\n{y_train.value_counts()}")

X_train shape: (455, 30)
y_train value counts:
```

```
target
1      285
0      170
Name: count, dtype: int64
```

✓ 3. Solution

Write your solution below. Submit **only** the `homework()` function to Omnicampus.

```
# Submit only this function to Omnicampus.
# Don't forget to erase the "!!WRITE ME!!" before submitting!
def homework(X_train, y_train, model_type, k):
    !!WRITE ME!!
    return mean_score
```

✓ 4. Test Your Solution

Run the test cell below to check that your `homework()` function does not contain import statements or data loading code before submitting.

Note: Do not modify the test code.

```
import ast
import inspect
import textwrap

def hw6_public_tests(homework_func):
    print("=== HW6 Public Tests ===")

    # Get the source code of the homework function
    source = textwrap.dedent(inspect.getsource(homework_func))
    tree = ast.parse(source)

    # ----- Test 1: No import statements -----
    print("Test 1: No import statements in function...", end=" ")
    for node in ast.walk(tree):
        if isinstance(node, (ast.Import, ast.ImportFrom)):
            print("NG")
            if isinstance(node, ast.Import):
                names = ", ".join(alias.name for alias in node.names)
            else:
                names = node.module
            print(f" Found import statement: '{names}'")
            print(" Remove all import statements before submitting.")
            return
    print("OK")
```

```

# ----- Test 2: No data loading code -----
print("Test 2: No data loading code in function...", end=" ")
blocked_calls = {"read_csv", "read_excel", "read_json", "read_html",
                  "get", "post", "ZipFile", "extractall", "urlopen"}
for node in ast.walk(tree):
    if isinstance(node, ast.Call):
        func_name = ""
        if isinstance(node.func, ast.Attribute):
            func_name = node.func.attr
        elif isinstance(node.func, ast.Name):
            func_name = node.func.id
        if func_name in blocked_calls:
            print("NG")
            print(f" Found data loading call: '{func_name}()'")
            print(" Do not include data downloading/loading code in your f
            return
print("OK")

# ----- Test 3: No hardcoded file paths or URLs -----
print("Test 3: No hardcoded file paths or URLs...", end=" ")
for node in ast.walk(tree):
    if isinstance(node, ast.Constant) and isinstance(node.value, str):
        val = node.value
        if val.startswith("http") or ".csv" in val or ".zip" in val:
            print("NG")
            print(f" Found hardcoded path/URL: '{val[:60]}...'")
            print(" Do not include file paths or URLs in your function.")
            return
print("OK")

print("=== All Public Tests Passed ===")

hw6_public_tests(homework)

```

✓ 5. Explore Your Results (Optional)

Try comparing the cross-validation scores of different models and different k values.

```

# Optional: Compare models and k values
models = ['logistic_regression', 'decision_tree', 'svm']
k_values = [3, 5, 10]

print(f"{'Model':<25} {'k':>3} {'Mean CV Accuracy':>18}")
print("-" * 50)
for model in models:
    for k in k_values:

```

```
score = homework(X_train, y_train, model, k)
print(f"{model:<25} {k:>3} {score:>18.6f}")
```

Once you've confirmed all public tests pass, copy and paste your `homework()` function to Omnicampus and submit.

If you see `1.0`, your answer is correct.

References

[1] Wolberg, W. *Breast Cancer Wisconsin (Diagnostic)*. UCI Machine Learning Repository. <https://doi.org/10.24432/C5DW2B>